A Self-Conjugate Partition Analog of $(t, t+1)$-Core Partitions with Distinct Parts

Huan Xiong and Lihong Yang*

Institute for Advanced Study in Mathematics
Harbin Institute of Technology
Heilongjiang 150001, P.R. China

Email: huan.xiong.math@gmail.com, yanglihong954@163.com

Abstract. Simultaneous core partitions have been extensively studied over the past two decades. In 2013, Amdeberhan proposed several conjectures regarding the number, the average size, and the largest size of $(t, t+1)$-core partitions with distinct parts. These conjectures were proved and generalized by Straub, Nath-Sellers, Zaleski-Zeilberger, Xiong, Paramonov, and many other mathematicians. In this paper, we introduce a natural self-conjugate partition analog of $(t, t+1)$-core partitions with distinct parts and derive their number, average size, and largest size.

Keywords. Integer partition, self-conjugate partition, hook length, core partition, average size, largest size.

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1 Introduction

A partition λ of a non-negative integer n is a finite non-increasing sequence of positive integers $(\lambda_1, \lambda_2, \ldots, \lambda_m)$ satisfying $\lambda_1 + \lambda_2 + \cdots + \lambda_m = n$ (see [19, 25]). We denote it by $\lambda \vdash n$. We say that λ_i $(1 \le i \le m)$ are the parts of λ and $|\lambda| := \sum_{1 \le i \le m} \lambda_i$ is the size of λ. The number m of parts of λ is called the length of λ, which is denoted by $\ell(\lambda)$. The *Young diagram* of λ is defined to be an up- and left-justified array of n boxes with λ_i boxes in the i-th row. The *conjugate partition* of λ is the partition whose Young diagram is obtained by reflecting the Young diagram of λ about the main diagonal, and λ is said to be *self-conjugate* if it is equal to its conjugate partition. For the (i, j)-box in the i-th row and j-th column in the Young diagram, its hook consists of the box itself together with all boxes to its right in the same row and all boxes below it in the same column. The *hook length* of this box is the total number of boxes contained in its hook. For example, Fig. 1 shows the Young diagram and hook lengths of the self-conjugate partition $(6, 4, 2, 2, 1, 1)$.

*Corresponding author.

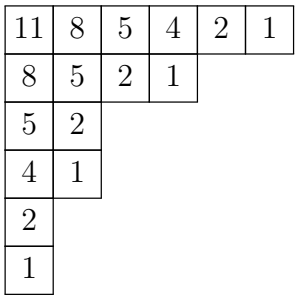


Fig. 1: The Young diagram and hook lengths of the self-conjugate partition $(6,4,2,2,1,1)$.

For a positive integer t, a partition is called a *t-core partition* if none of its hook lengths is divisible by t. Let t_i $(1 \le i \le k)$ be positive integers. A partition λ is a $(t_1, t_2, \ldots, t_k)$-core partition if it is simultaneously a t_1-core, a t_2-core, ..., and a t_k-core partition (see [1, 16]). For instance, as shown in Fig. 1, the partition $(6,4,2,2,1,1)$ is a $(6,7,13)$-core partition. Various statistics of simultaneous core partitions, such as their numbers, largest sizes, and average sizes, have been extensively studied over the past two decades (see [2, 3, 5–8, 10–15, 17, 18, 20–22, 26, 28–36]). For example, when s and t are coprime, Anderson [3] showed that the number of (s,t)-core partitions is equal to the rational Catalan number $\frac{1}{s+t}\binom{s+t}{s}$. In 2014, Armstrong, Hanusa, and Jones [5] conjectured that the average size of (s,t)-core partitions is equal to $\frac{(s+t-1)(s-1)(t-1)}{24}$ for coprime s and t. This conjecture was later proved independently by Johnson [15] and Wang [28].

A partition is self-conjugate if its Young diagram is symmetric along the main diagonal. For coprime s and t, Ford, Mai, and Sze [13] showed that the number of self-conjugate (s,t)-core partitions is $\binom{\lfloor \frac{s}{2} \rfloor + \lfloor \frac{t}{2} \rfloor}{\lfloor \frac{s}{2} \rfloor}$. Additionally, Armstrong, Hanusa, and Jones [5] conjectured in 2014 that their average size is $\frac{(s+t-1)(s-1)(t-1)}{24}$, a result later verified by Chen, Huang, and Wang [6] in 2016. Furthermore, Olsson and Stanton [23] proved that the largest size of such a partition is equal to $\frac{(s^2-1)(t^2-1)}{24}$.

In 2013, Amdeberhan [1] proposed conjectures on the number, average size, and largest size of $(t, t+1)$-core partitions with distinct parts. These were subsequently proved and generalized by Straub [27], Nath and Sellers [22], Zaleski and Zeilberger [34, 36], Xiong [29], Paramonov [24], and others.

Theorem 1.1 ([1, 27, 29]) *Let $t \ge 1$ be a given positive integer and $(F_i)_{i\ge 1} = (1,1,2,3,5,8,13,\ldots)$ denote the Fibonacci numbers. The following results hold for $(t, t+1)$-core partitions with distinct parts.*

(1) *The number of such partitions is F_{t+1}, where $F_t = F_{t-1} + F_{t-2}$ for $t \ge 3$, and the*

ordinary generating function is

$$\sum_{t\geq 0} F_{t+1}x^t = \frac{1}{1-x-x^2}. \tag{1.1}$$

(2) *The total sum of the sizes of these partitions and the average size are, respectively, given by*

$$\sum_{\substack{i+j+k=t+1\\ i,j,k\geq 1}} F_iF_jF_k \quad \text{and} \quad \sum_{\substack{i+j+k=t+1\\ i,j,k\geq 1}} \frac{F_iF_jF_k}{F_{t+1}}.$$

Moreover, the total sum H_t of the sizes of these partitions has the ordinary generating function

$$\sum_{t\geq 0} H_t x^t = \frac{x^2}{(1-x-x^2)^3}. \tag{1.2}$$

(3) *The largest size of such partitions is $\lfloor \frac{1}{3}\binom{t+1}{2} \rfloor$, where $\lfloor x \rfloor$ denotes the largest integer not greater than x.*

(4) *The number of partitions achieving the largest size is 2 if $t \equiv 1 \pmod 3$ and 1 otherwise.*

This paper aims to develop an appropriate analogue of Theorem 1.1 for self-conjugate partitions. A natural starting point is to consider self-conjugate $(t, t+1)$-core partitions with distinct parts. However, the structure of such partitions is straightforward, consisting precisely of $\{(n, n-1, n-2, \ldots, 3, 2, 1) : n \leq t/2\}$ (see [22]). Consequently, their number, average size, and largest size can be easily determined, making them an unsuitable analog for $(t, t+1)$-core partitions with distinct parts. Instead, we introduce and study the set $\mathcal{DS}(t)$, which consists of self-conjugate $(t, t+1)$-core partitions whose first $s(\lambda)$ parts are distinct, where $s(\lambda)$ is the size of the Durfee square of λ (see [4]). Recall that a partition λ has a Durfee square of size s if s is the largest integer such that λ has at least s parts each of size at least s (see [4]). We include the empty partition $\varnothing$ in $\mathcal{DS}(t)$. Studying such partitions is natural because a self-conjugate partition is uniquely determined by its first $s(\lambda)$ parts. For example, the partition $(6, 4, 2, 2, 1, 1) \in \mathcal{DS}(6)$ is a $(6, 7)$-core with $s(\lambda) = 2$, and its first two parts, 6 and 4, are distinct. Table 1 lists the partitions in $\mathcal{DS}(t)$ and their counts for $1 \leq t \leq 7$.

Table 1: Number of partitions in $\mathcal{DS}(t)$ and the set of such partitions.

t	*The set* $\mathcal{DS}(t)$	*The number* $f(t)$
1	$\{\varnothing\}$	1
2	$\{\varnothing, (1)\}$	2
3	$\{\varnothing, (1), (3,1,1)\}$	3
4	$\{\varnothing, (1), (2,1), (4,1,1,1)\}$	4
5	$\{\varnothing, (1), (2,1), (4,1,1,1), (5,1,1,1,1), (4,2,1,1)\}$	6
6	$\{\varnothing, (1), (2,1), (3,1,1), (3,2,1), (5,1,1,1,1), (5,2,1,1,1),$ $(6,1,1,1,1,1), (6,4,2,2,1,1)\}$	9
7	$\{\varnothing, (1), (2,1), (3,1,1), (3,2,1), (5,1,1,1,1), (5,2,1,1,1),$ $(5,3,2,1,1), (6,1,1,1,1,1), (6,2,1,1,1,1), (7,1,1,1,1,1,1),$ $(7,4,2,2,1,1,1), (7,6,2,2,2,2,1)\}$	13

For a partition λ, a main diagonal hook is the hook associated with a cell on the main diagonal of its Young diagram, and its length is the main diagonal hook length. Given a positive integer t, we adopt the following notation throughout this paper:

- $f(t)$ denotes the number of partitions in $\mathcal{DS}(t)$,
- $g(t)$ denotes the total number of main diagonal hooks among all partitions in $\mathcal{DS}(t)$,
- $h(t)$ denotes the sum of the sizes of all partitions in $\mathcal{DS}(t)$.

We derive the following results for the partitions in $\mathcal{DS}(t)$.

Theorem 1.2 *Let $t \geq 1$ be an integer. The following results hold for $f(t)$, $g(t)$, and $h(t)$ associated with self-conjugate partitions in $\mathcal{DS}(t)$.*

(1) *Let $f(-1) = f(0) = 1$. Then $f(1) = 1$ and $f(2) = 2$. For $t \geq 3$, we have $f(t) = f(t-1) + f(t-3)$. Consequently, the ordinary generating function for $f(t-1)$ is*

$$\sum_{t\geq 0} f(t-1)x^t = \frac{1}{1-x-x^3}. \tag{1.3}$$

(2) *Let $g(-1) = g(0) = 0$. Then $g(1) = 0$, $g(2) = 1$, and the ordinary generating function for $g(t-1)$ is*

$$\sum_{t\geq 0} g(t-1)x^t = \frac{x^3}{(1-x-x^3)^2}. \tag{1.4}$$

(3) *Let* $h(-1) = h(0) = 0$. *Then* $h(1) = 0$, $h(2) = 1$, *and the ordinary generating function for* $h(t-1)$ *is*

$$\sum_{t\geq 0} h(t-1)x^t = \frac{(1+4x-x^2+6x^3+4x^4+5x^6)x^3}{(1+x+x^3)(1-x-x^3)^3}. \tag{1.5}$$

Theorem 1.3 *Let* t *be an integer and let* $m(t)$ *denote the largest size of a self-conjugate* $(t, t+1)$*-core partition in* $\mathcal{DS}(t)$. *Then* $m(1) = 0$, $m(2) = 1$, *and* $m(t) = 2t-1$ *for* $3 \leq t \leq 5$. *For* $t \geq 6$, *the following results hold:*

(1) *When* (i) t *is even and* $\frac{t}{2} - 1 - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 0 \pmod 3$*; or* (ii) t *is odd and* $\frac{t-1}{2} - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 2 \pmod 3$, *then*

$$m(t) = \frac{1}{3}(t-1)\left(t+1-4\lfloor \frac{2t+1}{12} \rfloor\right) + \lfloor \frac{2t+1}{12} \rfloor \left(2t+1-2\lfloor \frac{2t+1}{12} \rfloor\right).$$

(2) *When* (iii) t *is even and* $\frac{t}{2} - 1 - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 1 \pmod 3$*; or* (iv) t *is odd and* $\frac{t-1}{2} - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 0 \pmod 3$, *then*

$$m(t) = \frac{1}{3}(t-1)\left(t-7-4\lfloor \frac{2t+1}{12} \rfloor\right) + \left(\lfloor \frac{2t+1}{12} \rfloor + 2\right)\left(2t-3-2\lfloor \frac{2t+1}{12} \rfloor\right).$$

(3) *When* (v) t *is even and* $\frac{t}{2} - 1 - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 2 \pmod 3$*; or* (vi) t *is odd and* $\frac{t-1}{2} - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 1 \pmod 3$, *then*

$$m(t) = \frac{1}{3}(t-1)\left(t-3-4\lfloor \frac{2t+1}{12} \rfloor\right) + \left(\lfloor \frac{2t+1}{12} \rfloor + 1\right)\left(2t-1-2\lfloor \frac{2t+1}{12} \rfloor\right).$$

(4) *There is always a unique partition in* $\mathcal{DS}(t)$ *achieving the largest size.*

Here $\lfloor x \rfloor$ *denotes the largest integer not greater than* x.

Remark 1.4 *From the above results, we observe that the number* $f(t)$ *of partitions in* $\mathcal{DS}(t)$ *satisfies the recurrence* $f(t) = f(t-1) + f(t-3)$, *while the number* F_{t+1} *of* $(t, t+1)$*-core partitions with distinct parts satisfies* $F_t = F_{t-1} + F_{t-2}$. *Thus, these two families of partitions exhibit analogous recurrence structures. Furthermore, their largest sizes are both of order* $O(t^2)$, *and deriving exact formulas requires case analysis based on the residue of* t *modulo certain integers. Regarding the average size, we focus on the total sum of sizes. The generating functions (1.2) and (1.5) for these total sums can be factored as the cube of the generating function for the counts multiplied by a rational function. Therefore, partitions in* $\mathcal{DS}(t)$ *can be viewed as a natural self-conjugate analog of* $(t, t+1)$*-core partitions with distinct parts.*

Let λ be a partition. Define $MD(\lambda) = \{h_1, h_2, \ldots, h_k\}$ to be the set of its main diagonal hook lengths, where $h_1 > h_2 > \cdots > h_k$ are the hook lengths of the cells lying on the main diagonal of λ. The number of such hook lengths is denoted by $k = s(\lambda) = |MD(\lambda)|$, with $MD(\varnothing) = \varnothing$. Let $d(\lambda)$ denote the sequence of consecutive differences of these main diagonal hook lengths, where $d_i = h_i - h_{i+1}$ for $1 \le i \le k-1$. For example, if $\lambda = (5,3,2,1,1)$, then $MD(\lambda) = \{9,3\}$ and $d(\lambda) = (d_1) = (6)$. For a nonnegative integer m, we say $d(\lambda) \ge m$ if $d_i \ge m$ for all $1 \le i \le |MD(\lambda)| - 1$. A self-conjugate $(t, t+1)$-core partition λ belongs to $\mathcal{DS}(t)$ if and only if $d(\lambda) \ge 4$ (when $k = 1$, $d(\lambda)$ is empty and thus automatically satisfies $d(\lambda) \ge 4$). It is well known that a self-conjugate partition is uniquely determined by its main diagonal hook lengths, which are distinct odd numbers. Henceforth, we identify a self-conjugate partition λ with its set of main diagonal hook lengths. Table 2 lists the largest sizes of partitions in $\mathcal{DS}(t)$ and their $MD(\lambda)$ for $3 \le t \le 26$.

Table 2: Largest size of partitions in $\mathcal{DS}(t)$ and their main diagonal hook lengths.

t	$MD(\lambda)$	*Largest size*	t	$MD(\lambda)$	*Largest size*
3	$\{5\}$	5	15	$\{29,25,21,17\}$	92
4	$\{7\}$	7	16	$\{31,27,21,15,9\}$	103
5	$\{9\}$	9	17	$\{33,29,25,19,13\}$	119
6	$\{11,5\}$	16	18	$\{35,31,27,23,17\}$	133
7	$\{13,9\}$	22	19	$\{37,33,29,25,21\}$	145
8	$\{15,11\}$	26	20	$\{39,35,31,25,19,13\}$	162
9	$\{17,11,5\}$	33	21	$\{41,37,33,29,23,17\}$	180
10	$\{19,15,9\}$	43	22	$\{43,39,35,31,27,21\}$	196
11	$\{21,17,13\}$	51	23	$\{45,41,37,31,25,19,13\}$	211
12	$\{23,19,15\}$	57	24	$\{47,43,39,35,29,23,17\}$	233
13	$\{25,21,15,9\}$	70	25	$\{49,45,41,37,33,27,21\}$	253
14	$\{27,23,19,13\}$	82	26	$\{51,47,43,39,35,31,25\}$	271

The rest of this paper is organized as follows. Section 2 proves the enumeration results for partitions in $\mathcal{DS}(t)$. Section 3 presents the proof of the largest size results.

2 Proof of Theorem 1.2

In this section, we will prove Theorem 1.2 by deriving the recurrence relations of $f(t)$, $g(t)$, and $h(t)$. We begin by recalling a result of Ford, Mai, and Sze [13].

Theorem 2.1 ([13]) *Let λ be a self-conjugate partition. Then λ is a t-core partition if and only if the following conditions hold:*

(i) *If $h \in MD(\lambda)$ and $h > 2t$, then $h - 2t \in MD(\lambda)$,*

(ii) *If $h_1, h_2 \in MD(\lambda)$, then $h_1 + h_2 \not\equiv 0 \pmod{2t}$.*

From the definition of main diagonal hook lengths, we directly obtain the following result.

Lemma 2.2 *The size of a self-conjugate partition equals the sum of its main diagonal hook lengths. That is, if λ is self-conjugate, then*

$$|\lambda| = \sum_{x \in MD(\lambda)} x.$$

The next lemma provides an upper bound for the main diagonal hook lengths of a partition in $\mathcal{DS}(t)$.

Lemma 2.3 *Let t be a positive integer and $\lambda \in \mathcal{DS}(t)$. If $h \in MD(\lambda)$, then $h \leq 2t-1$.*

Proof. Suppose, for contradiction, that there exists $h \in MD(\lambda)$ with $h > 2t - 1$. Since λ is a $(t, t+1)$-core, h cannot be $2t$ or $2t+2$. If $h = 2t+1 > 2t$, then by Theorem 2.1(i), we have $1 \in MD(\lambda)$. However, by Theorem 2.1(ii), this is impossible because $1 + (2t+1) = 2(t+1)$. Furthermore, if $h \geq 2t+3$, then by Theorem 2.1(i), $h - 2t, h - (2t+2) \in MD(\lambda)$. Suppose $h - 2t$ is the s-th main diagonal hook length of λ, where $1 \leq s \leq |MD(\lambda)| - 2$. Then $h - (2t+2)$ is the $(s+1)$-st main diagonal hook length. Recall that $d(\lambda)$ denotes the sequence $(d_1, d_2, \ldots, d_{|MD(\lambda)|-1})$, where d_i is the difference between the i-th and the $(i+1)$-th main diagonal hook lengths of λ. Then $d_s = h_s - h_{s+1} = (h - 2t) - (h - (2t+2)) = 2$, contradicting the assumption that $\lambda \in \mathcal{DS}(t)$ (which requires $d(\lambda) \geq 4$). Hence, $h \leq 2t - 1$ for all $h \in MD(\lambda)$. ∎

For a positive integer t, define $\mathrm{Q}_t =: \{1, 3, 5, 7, \ldots, 2t-1\} \backslash \{t, t+1\}$. Clearly, Q_t consists of distinct positive odd integers. For example, when $t = 12$, $\mathrm{Q}_{12} = \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23\}$.

Combining Theorem 2.1 and Lemma 2.3, we obtain the following characterization of $MD(\lambda)$ for $\lambda \in \mathcal{DS}(t)$.

Theorem 2.4 *Let t be a positive integer. Then a self-conjugate partition λ belongs to $\mathcal{DS}(t)$ if and only if $MD(\lambda)$ satisfies:*

(i) $MD(\lambda) \subseteq Q_t$,

(ii) *If* $h_1, h_2 \in MD(\lambda)$ *and* $h_1 > h_2$, *then* $h_1 - h_2 \geq 4$,

(iii) *If* $h_1, h_2 \in MD(\lambda)$, *then* $h_1 + h_2 \neq 2t$ *and* $h_1 + h_2 \neq 2t + 2$.

The following lemma is essential for proving Theorem 1.2.

Lemma 2.5 *For* $t \geq 3$, *the following recurrence relations hold:*

$$f(t) = f(t-1) + f(t-3), \tag{2.1}$$
$$g(t) = g(t-1) + g(t-3) + f(t-3), \tag{2.2}$$
$$h(t) = 2tg(t-1) + (2t-2)g(t-3) - h(t-1) - h(t-3) + f(t-3), \tag{2.3}$$

with initial values $g(0) = h(0) = g(1) = h(1) = 0$, $f(0) = f(1) = g(2) = h(2) = 1$, *and* $f(2) = 2$.

Proof. Let $\lambda \in \mathcal{DS}(t)$ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$ where $x_i > x_{i+1}$ for $1 \leq i \leq m-1$. From the definitions of $f(t)$, $g(t)$, $h(t)$, and Lemma 2.2, we have:

$$f(t) = \sum_{\lambda \in \mathcal{DS}(t)} 1, \quad g(t) = \sum_{\lambda \in \mathcal{DS}(t)} |MD(\lambda)|, \quad \text{and } h(t) = \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ x \in MD(\lambda)}} x. \tag{2.4}$$

We proceed by considering two cases.

Case 1. Suppose that $1 \notin MD(\lambda)$. Let $\Omega(t)$ be the set of all $\lambda \in \mathcal{DS}(t)$ with $MD(\lambda) \subseteq Q_t \backslash \{1\}$. Define a map $\psi_t : \Omega(t) \to \mathcal{DS}(t-1)$ as follows. If $\lambda = \varnothing$, then set $\psi_t(\lambda) = \varnothing$. If $\lambda \neq \varnothing$, then define $\psi_t(\lambda)$ as the partition μ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, where $y_i = 2t - x_{m+1-i}$ for $1 \leq i \leq m$.

Claim 1. $MD(\mu) \subseteq Q_{t-1}$.
By the definition of the map ψ_t, for each $y_i \in MD(\mu)$ we have $y_i = 2t - x_{m+1-i}$. Since x_{m+1-i} is odd and satisfies $3 \leq x_{m+1-i} \leq 2t-1$, with $x_{m+1-i} \neq t, t+1$, it follows that y_i is odd, $1 \leq y_i \leq 2t-3$, and $y_i \neq t-1, t$. Thus, by the definition of Q_{t-1}, we conclude that $y_i \in Q_{t-1}$, and thus $MD(\mu) \subseteq Q_{t-1}$, as claimed.

Claim 2. $d(\mu) \geq 4$.
Otherwise, there exist $i < j$ such that $y_i, y_j \in MD(\mu)$ and $y_i - y_j < 4$. Then from the definition of the map ψ_t, we have $2t - x_{m+1-i} > 2t - x_{m+1-j}$ and $(2t - x_{m+1-i}) - (2t - x_{m+1-j}) < 4$. Equivalently, $x_{m+1-j} > x_{m+1-i}$ and $x_{m+1-j} - x_{m+1-i} < 4$. This contradicts $\lambda \in \mathcal{DS}(t)$. Therefore, $y_i - y_j \geq 4$ for all $1 \leq i < j \leq m$, that is, $d(\mu) \geq 4$.

Claim 3. $y_i + y_j \neq 2t-2$ and $y_i + y_j \neq 2t$ for any $i, j \in [1, m]$ and $i \neq j$.
Otherwise, there exist $i < j$ such that $y_i, y_j \in MD(\mu)$ and $y_i + y_j$ equals either $2t-2$

or $2t$. If $y_i + y_j = 2t-2$, then $(2t - x_{m+1-i}) + (2t - x_{m+1-j}) = 2t-2$, which implies that $x_{m+1-i} + x_{m+1-j} = 2t+2$, contradicting Theorem 2.4(iii). The argument for the case $y_i + y_j \neq 2t$ is similar and therefore omitted. Hence, the claim is proved.

For example, let $t = 8$ and $\lambda = (7,5,2,2,2,1) \in \mathcal{DS}(8)$ with $MD(\lambda) = \{13,7\}$. Then $MD(\psi_8(\lambda)) = \{9,3\}$ as shown in Fig. 2, that is, $\psi_8(\lambda) = (5,3,2,1,1) \in \mathcal{DS}(7)$.

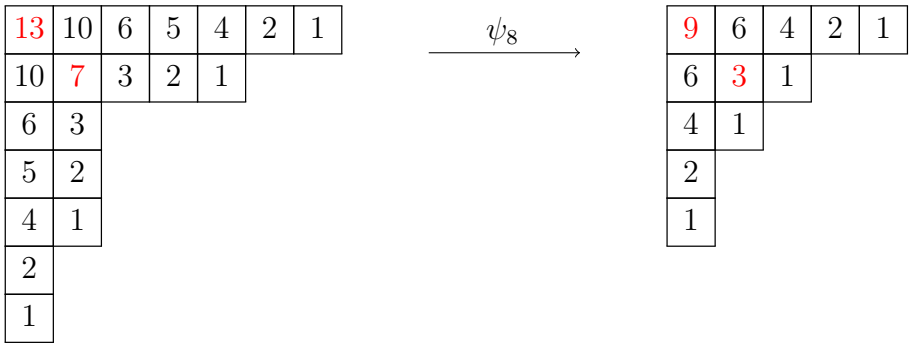


Fig. 2: An example of Case 1.

Thus, $MD(\mu)$ satisfies the properties in Theorem 2.4, so $\mu \in \mathcal{DS}(t-1)$, and ψ_t is well-defined. Conversely, given any $\mu \in \mathcal{DS}(t-1)$, one can recover $\lambda \in \Omega(t)$ by setting $MD(\lambda) = \{2t - y_{m+1-i} : 1 \leq i \leq m\}$ if $\mu \neq \varnothing$, and $\lambda = \varnothing$ if $\mu = \varnothing$. This shows ψ_t is a bijection. Moreover, it is straightforward to verify that $MD(\lambda)$ satisfies the three properties stated in Theorem 2.4. Consequently, $\lambda \in \mathcal{DS}(t)$ with $1 \notin MD(\lambda)$.

Case 2. Suppose $1 \in MD(\lambda)$. From Theorem 2.4 (ii) and (iii), we have $3, 2t-1 \notin MD(\lambda)$ because $3-1 = 2 < 4$ and $(2t-1)+1 = 2t$. Let $\Omega'(t)$ be the set of partitions λ' obtained from λ by deleting the main diagonal hook length 1, i.e., $MD(\lambda') = MD(\lambda)\backslash\{1\} = \{x_1, x_2, \ldots, x_{m-1}\}$. Then $MD(\lambda') \subseteq \mathrm{Q}_t\backslash\{1,3,2t-1\}$. Define a map $\phi_t : \Omega'(t) \to \mathcal{DS}(t-3)$ as follows. If $\lambda' = \varnothing$(i.e., $\lambda = (1)$), set $\phi_t(\lambda') = \varnothing$. If $\lambda' \neq \varnothing$, define $\phi_t(\lambda')$ as the partition μ with $MD(\mu) = \{y_1, y_2, \ldots, y_{m-1}\}$, where $y_i = 2t-2-x_{m-i}$ for $1 \leq i \leq m-1$.

Claim 1. $MD(\mu) \subseteq \mathrm{Q}_{t-3}$.
By the definition of the map ψ_t, for each $y_i \in MD(\mu)$ we have $y_i = 2t-2-x_{m-i}$. Since x_{m-i} is odd and satisfies $5 \leq x_{m-i} \leq 2t-3$ with $x_{m-i} \neq t, t+1$, it follows that y_i is odd, $1 \leq y_i \leq 2t-7$, and $y_i \neq t-3, t-2$. Thus, $y_i \in \mathrm{Q}_{t-3}$ and therefore $MD(\mu) \subseteq \mathrm{Q}_{t-3}$, as claimed.

Claim 2. $d(\mu) \geq 4$.
Otherwise, there exist $i < j$ such that such that $y_i, y_j \in MD(\mu)$ and $y_i - y_j < 4$. Then $2t-2-x_{m-i} > 2t-2-x_{m-j}$ and $(2t-2-x_{m-i}) - (2t-2-x_{m-j}) < 4$. Equivalently, $x_{m-j} > x_{m-i}$ and $x_{m-j} - x_{m-i} < 4$. Since $x_{m-j}, x_{m-i} \in MD(\lambda')$ and $MD(\lambda') = MD(\lambda)\backslash\{1\}$, we obtain $x_{m-j}, x_{m-i} \in MD(\lambda)$. This contradicts $\lambda \in \mathcal{DS}(t)$. Hence, $y_i - y_j \geq 4$ for

all $1 \le i < j \le m-1$, i.e., $d(\mu) \ge 4$.

Claim 3. $y_i + y_j \ne 2t-6$ and $y_i + y_j \ne 2t-4$ for any $i \ne j$.
Otherwise, there exist indices $i < j$ such that $y_i, y_j \in MD(\mu)$ and $y_i + y_j$ equals either $2t-6$ or $2t-4$. If $y_i + y_j = 2t-6$, then $(2t-2-x_{m-i}) + (2t-2-x_{m-j}) = 2t-6$, so $x_{m-i} + x_{m-j} = 2t+2$. Since $x_{m-j}, x_{m-i} \in MD(\lambda')$ and $MD(\lambda') = MD(\lambda)\backslash\{1\}$, we obtain $x_{m-j}, x_{m-i} \in MD(\lambda)$. This contradicts Theorem 2.4(iii). The proof for the case $y_i + y_j \ne 2t-4$ is similar and therefore omitted. Hence, the claim follows.

For example, let $t = 10$ and $\lambda = (5,4,3,2,1) \in \mathcal{DS}(10)$ with $MD(\lambda) = \{9,5,1\}$. Then $\lambda' = (5,4,2,2,1)$ with $MD(\lambda') = \{9,5\}$. Applying ϕ_{10} gives $MD(\phi_{10}(\lambda')) = \{13,9\}$, as shown in Fig. 3. Therefore, $\phi_{10}(\lambda') = (7,6,2,2,2,2,1) \in \mathcal{DS}(7)$.

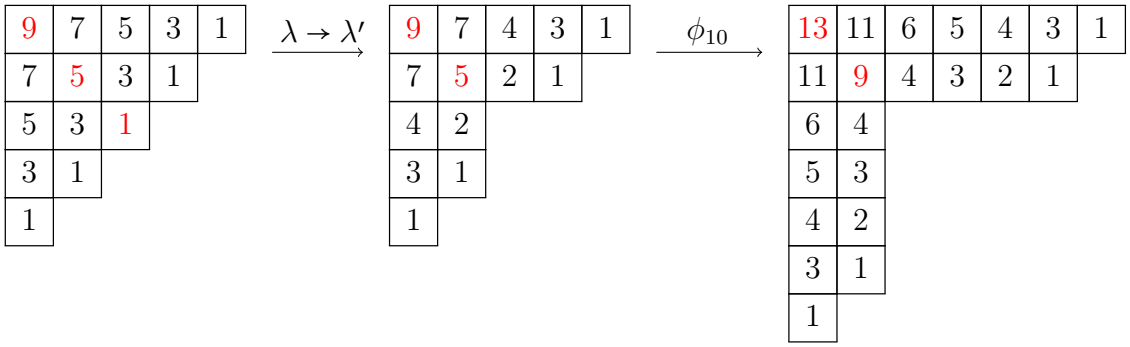


Fig. 3: An example of Case 2.

Therefore, $MD(\mu)$ satisfies the three properties in Theorem 2.4. It follows that $\mu \in \mathcal{DS}(t-3)$, and consequently, the map ϕ_t is well-defined. Conversely, given $\mu \in \mathcal{DS}(t-3)$, one can recover the partition $\lambda' \in \Omega'(t)$ by setting $MD(\lambda') = \{2t-2-y_{m+1-i} : 1 \le i \le m\}$ if $\mu \ne \varnothing$, and $\lambda' = \varnothing$ otherwise. Then λ with $MD(\lambda) = MD(\lambda') \cup \{1\}$ belongs to $\mathcal{DS}(t)$ with $1 \in MD(\lambda)$. Hence, ϕ_t is a bijection. Furthermore, let λ be the partition obtained from λ' by adding 1 to the set of main diagonal hook lengths: $MD(\lambda) = MD(\lambda') \cup \{1\} = \{x_1, x_2, \ldots, x_m, 1\}$. It is straightforward to verify that $MD(\lambda)$ satisfies the three properties in Theorem 2.4, which implies that $\lambda \in \mathcal{DS}(t)$ with $1 \in MD(\lambda)$.

We now proceed to prove (2.1), (2.2) and (2.3). By combining the two cases discussed above with (2.4), we immediately obtain the following results:

$$f(t) = \sum_{\lambda \in \mathcal{DS}(t)} 1 = \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ 1 \in MD(\lambda)}} 1 + \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ 1 \notin MD(\lambda)}} 1 = \sum_{\lambda \in \mathcal{DS}(t-3)} 1 + \sum_{\lambda \in \mathcal{DS}(t-1)} 1 = f(t-3) + f(t-1)$$

and

$$g(t) = \sum_{\lambda \in \mathcal{DS}(t)} |MD(\lambda)| = \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ 1 \in MD(\lambda)}} |MD(\lambda)| + \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ 1 \notin MD(\lambda)}} |MD(\lambda)|$$
$$= \sum_{\lambda \in \Omega'(t)} (|MD(\lambda)| + 1) + \sum_{\lambda \in \Omega(t)} |MD(\lambda)| = g(t-3) + f(t-3) + g(t-1).$$

Similarly, it is routine to check that

$$h(t) = \sum_{\lambda \in \mathcal{DS}(t)} |\lambda| = \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ 1 \in MD(\lambda)}} |\lambda| + \sum_{\substack{\lambda \in \mathcal{DS}(t) \\ 1 \notin MD(\lambda)}} |\lambda| = \sum_{\lambda \in \Omega'(t)} (|\lambda| + 1) + \sum_{\lambda \in \Omega(t)} |\lambda|$$
$$= \sum_{\substack{\lambda \in \mathcal{DS}(t-3) \\ x \in MD(\lambda)}} (2t - 2 - x) + \sum_{\lambda \in \mathcal{DS}(t-3)} 1 + \sum_{\substack{\lambda \in \mathcal{DS}(t-1) \\ x \in MD(\lambda)}} (2t - x)$$
$$= 2tg(t-1) + (2t-2)g(t-3) - h(t-1) - h(t-3) + f(t-3)$$

as desired, completing the proof. ∎

Now we are ready to prove Theorem 1.2.

Proof of Theorem 1.2. Let

$$F(x) = \sum_{t \geq 0} f(t-1)x^t, \quad G(x) = \sum_{t \geq 0} g(t-1)x^t, \quad \text{and} \quad H(x) = \sum_{t \geq 0} h(t-1)x^t.$$

We set $f(-1) = 1$ and $g(-1) = h(-1) = 0$. From the recurrence relation (2.1) and initial values $f(-1) = f(0) = f(1) = 1$ and $f(2) = 2$, we have

$$F(x) = 1 + xF(x) + x^3F(x),$$

which implies (1.3) as desired.

Similarly, from the recurrence relation (2.2) and initial values, we deduce that

$$G(x) = xG(x) + x^3G(x) + x^3F(x),$$

which implies

$$G(x) = \frac{x^3F(x)}{1 - x - x^3}. \tag{2.5}$$

Next, substituting (1.3) into (2.5), we derive (1.4) as desired.

Furthermore, from the recurrence relation (2.3) and initial values, we derive

$$\frac{H(x) - x^3}{x} = 2xG'(x) + 2(x^2G(x))' - 2x^2G(x) - H(x) - x^2H(x) + x^2F(x) - x^2. \tag{2.6}$$

Simplifying (2.6) gives

$$H(x) = 2x^3G(x) + (2x^2 + 2x^4)G'(x) - xH(x) - x^3H(x) + x^3F(x),$$

which implies

$$H(x) = \frac{2x^3G(x) + (2x^2 + 2x^4)G'(x) + x^3F(x)}{1 + x + x^3}. \tag{2.7}$$

Differentiating (1.4) gives:

$$G'(x) = \frac{x^2(3 - x + 3x^3)}{(1 - x - x^3)^3}. \tag{2.8}$$

Substituting (1.3), (1.4), and (2.8) into (2.7) yields (1.5), completing the proof. ∎

3 Proof of Theorem 1.3

This section is devoted to proving Theorem 1.3. We will characterize the partitions in $\mathcal{DS}(t)$ achieving the largest size through a series of lemmas.

Lemma 3.1 *For an integer $t \geq 3$, let λ be a partition in $\mathcal{DS}(t)$ achieving the largest size. Then $2t-1 \in MD(\lambda)$. Furthermore, $1, 3, 2t-3 \notin MD(\lambda)$.*

Proof. Suppose $\lambda \in \mathcal{DS}(t)$ achieves the largest size. Let $MD(\lambda) = MD'(\lambda) \cup MD''(\lambda)$, where $MD'(\lambda) \subseteq Q_t \backslash \{1, 3, 2t-3, 2t-1\}$ and $MD''(\lambda) \subseteq \{1, 3, 2t-3, 2t-1\}$. Assume, for contradiction, that $MD''(\lambda) \subseteq \{1, 3, 2t-3\}$. Since $3 + (2t-3) = 2t$, by Theorem 2.4(iii), 3 and $2t-3$ cannot both be in $MD''(\lambda)$. Since $t \geq 3$, we have $2t - 3 \geq 3$. Hence, the maximal possible sum of elements in $MD''(\lambda)$ is at most $1 + (2t-3)$, i.e.

$$\sum_{x \in MD''(\lambda)} x \leq 1 + (2t-3).$$

Let μ be a self-conjugate partition with $MD(\mu) = MD'(\lambda) \cup \{2t-1\}$.

Since $MD(\mu)$ satisfies the three conditions in Theorem 2.4, it follows that $\mu \in \mathcal{DS}(t)$. By Lemma 2.2 and the bound above,

$$|\lambda| \leq (1 + (2t-3)) + \sum_{x \in MD'(\lambda)} x < (2t-1) + \sum_{x \in MD'(\lambda)} x = |\mu|.$$

contradicting the maximality of λ. Hence, $2t-1 \in MD(\lambda)$. By Theorem 2.4(ii) and (iii), we have $1, 3, 2t-3 \notin MD(\lambda)$ because $1 + (2t-1) = 2t$, $3 + (2t-1) = 2(t+1)$, and $(2t-1) - (2t-3) = 2 < 4$, completing the proof. ∎

Define $Q_t^l = Q_t \cap [1, t-1]$ and $Q_t^r = Q_t \cap [t+2, 2t-1]$. Then $Q_t = Q_t^l \cup Q_t^r$. For example, when $t = 12$, we have $Q_{12}^l = \{1, 3, 5, 7, 9, 11\}$ and $Q_{12}^r = \{15, 17, 19, 21, 23\}$. Fig. 4 illustrates Q_{12}^l and Q_{12}^r, where two numbers are connected if their sum is $2t$ or $2t+2$.

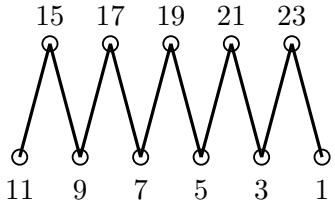


Fig. 4: The representation of Q_{12}^l and Q_{12}^r.

We now describe the sequence $d(\lambda)$ for a maximal partition $\lambda \in \mathcal{DS}(t)$. Note that the sequence $d(\lambda)$ is nonempty only if $t \geq 6$. For $t \geq 6$, we construct a partition $\lambda' \in \mathcal{DS}(t)$ with $|\lambda'| > |\lambda|$ to derive contradictions in the following lemmas.

Lemma 3.2 *For an integer $t \geq 6$, let $\lambda \in \mathcal{DS}(t)$ have the largest size. If $h_i, h_{i+1} \in MD(\lambda)$ for $1 \leq i \leq |MD(\lambda)| - 1$, then $4 \leq d_i \leq 6$, i.e., $d_i = 4$ or 6.*

Proof. Suppose $\lambda \in \mathcal{DS}(t)$ achieves the largest size, where $t \geq 6$. By Theorem 2.4(i), all $h_i \in MD(\lambda)$ are odd. Assume, for contradiction, that there exist consecutive h_j and h_{j+1} such that $d_j \geq 8$, that is, $h_j - h_{j+1} \geq 8$. Let j be the smallest such index. We consider the following cases.

Case 1. $h_j, h_{j+1} \in Q_t^r$.
Subcase 1.1. If $2t+4-h_j$ (resp. $2t+6-h_j$) $\in MD(\lambda)$, define λ' by replacing $2t+4-h_j$ (resp., $2t+6-h_j$) with $h_j - 4$ in $MD(\lambda)$.
Subcase 1.2. If neither $2t+4-h_j$ nor $2t+6-h_j$ is in $MD(\lambda)$, define λ' by setting $MD(\lambda') = MD(\lambda) \cup \{h_j - 4\}$.

For example, let $\lambda \in \mathcal{DS}(16)$ with $MD(\lambda) = \{31, 23, 19, 5\}$. Then $d(\lambda) = (8, 4, 14)$ and $h_1 = 31, h_2 = 23 \in Q_{16}^r$. Moreover, since $2 \times 16 + 4 - 31 = 5 \in MD(\lambda)$, it follows from the construction in Subcase 1.1 that we obtain a new partition $\lambda' \in \mathcal{DS}(16)$ with $MD(\lambda') = \{31, 27, 23, 19\}$.

Case 2. $h_j \in Q_t^r$ and $h_{j+1} \in Q_t^l$. Then $2t - h_{j+1} \notin MD(\lambda)$.
Subcase 2.1. When t is even, $t+3$ is the smallest element in Q_t^r. We consider the following possibilities.

(1) If $h_j = t+3$, define λ' by replacing every $x \in MD(\lambda)$ with $h_j \leq x \leq 2t-2-h_{j+1}$ by $x+2$, and replacing h_{j+1} by $t-1$.

(2) If $h_j \neq t+3$ and $h_{j+1} \notin \{t-1, t-3\}$, define λ' by setting $MD(\lambda') = MD(\lambda) \cup \{t-1\}$.

(3) If $h_j \neq t+3$ and $h_{j+1} \in \{t-1, t-3\}$, then $t+5 \notin MD(\lambda)$. Hence, define λ' by replacing h_{j+1} with $t+3$ in $MD(\lambda)$.

For example, let $\lambda \in \mathcal{DS}(18)$ with $MD(\lambda) = \{35, 31, 25, 21, 9\}$. Then $d(\lambda) = (4, 6, 4, 12)$ and $h_4 = 18 + 3 = 21 \in \mathrm{Q}_{18}^r$, $h_5 = 9 \in \mathrm{Q}_{18}^l$. By the above procedure in Subcase 2.1(1), we obtain $\lambda' \in \mathcal{DS}(18)$ with $MD(\lambda') = \{35, 31, 27, 23, 17\}$.

Subcase 2.2. When t is odd, the smallest element of Q_t^r is $t+2$. We consider the following cases.

(1) If $h_j = t+2$, define λ' by replacing every $x \in MD(\lambda)$ with $h_j+4 \leq x \leq 2t-2-h_{j+1}$ by $x+2$, and replacing h_{j+1} by $t-4$.

(2) If $h_j \neq t+2$ and $h_j < 2t - h_{j+1}$, define λ' by replacing every $x \in MD(\lambda)$ with $h_j \leq x \leq 2t-2-h_{j+1}$ by $x+2$, and replacing h_{j+1} by $t+2$.

(3) If $h_j \neq t+2$ and $h_j > 2t - h_{j+1}$, define λ' by replacing h_{j+1} with $t+2$ in $MD(\lambda)$.

For example, let $\lambda \in \mathcal{DS}(15)$ with $MD(\lambda) = \{29, 25, 19, 9\}$. Then we have $d(\lambda) = (4, 6, 10)$, $h_3 = 19 \in \mathrm{Q}_{15}^r$, $h_4 = 9 \in \mathrm{Q}_{15}^l$, and $h_3 = 19 < 2 \times 15 - h_4 = 21$. By the above procedure in Subcase 2.2(2), we obtain λ' with $MD(\lambda') = \{29, 25, 21, 17\}$.

Case 3. $h_j, h_{j+1} \in \mathrm{Q}_t^l$. Then $2t - h_{j+1}, 2t + 2 - h_{j+1} \notin MD(\lambda)$. By Lemma 3.1 and $h_j - h_{j+1} \geq 8$, we obtain $5 \leq h_{j+1} \leq t-9$. By the minimality of j, we have $4 \leq d_i \leq 6$ for all $i < j$. Since $2t - h_{j+1}, 2t + 2 - h_{j+1} \notin MD(\lambda)$, we derive $2t + 4 - h_{j+1}, 2t - 2 - h_{j+1} \in MD(\lambda)$. Furthermore, since $2t + 4 - h_{j+1} \in MD(\lambda)$, we have $h_{j+1} + 4 \notin MD(\lambda)$. Construct a new partition λ' by replacing $2t - 2 - h_{j+1}$ with $2t - h_{j+1}$ and h_{j+1} with $h_{j+1} + 4$ in $MD(\lambda)$.

For example, let $\lambda \in \mathcal{DS}(19)$ with $MD(\lambda) = \{37, 31, 27, 21, 15, 5\}$. Then we have $d(\lambda) = (6, 4, 6, 6, 10)$ and $h_5 = 15, h_6 = 5 \in \mathrm{Q}_{19}^l$. By the above procedure in Case 3, we obtain $\lambda' \in \mathcal{DS}(19)$ with $MD(\lambda') = \{37, 33, 27, 21, 15, 9\}$.

In all cases, it is straightforward to verify that $MD(\lambda')$ satisfies the three conditions stated in Theorem 2.4, and the sum of its elements is greater than that of $MD(\lambda)$. By Lemma 2.2, $|\lambda'| > |\lambda|$, contradicting the maximality of λ. Hence, $d_i \in \{4, 6\}$. ∎

Lemma 3.3 *For $t \geq 6$, let $\lambda \in \mathcal{DS}(t)$ achieve the largest size. If $h_i, h_{i+1} \in MD(\lambda) \cap \mathrm{Q}_t^r$ and $d_i = 6$ for some $1 \leq i \leq |MD(\lambda)| - 1$, then $2t + 4 - h_i \in MD(\lambda)$.*

Proof. Assume to the contrary that the lemma does not hold. Then there exist $h_i, h_{i+1} \in MD(\lambda) \cap \mathrm{Q}_t^r$ with $d_i = 6$ (that is, $h_i = h_{i+1} + 6$) and $2t + 4 - h_i = 2t - 2 - h_{i+1} \notin MD(\lambda)$. Consequently, $2t + 6 - h_i = 2t - h_{i+1} \notin MD(\lambda)$ and $2t + 2 - h_i \notin MD(\lambda)$. We then construct a new partition λ' by setting $MD(\lambda') = MD(\lambda) \cup \{2t + 4 - h_i\}$. It is straightforward to verify that $MD(\lambda')$ satisfies the three conditions stated in Theorem 2.4 and has a larger element sum than $MD(\lambda)$, thus $|\lambda'| > |\lambda|$, a contradiction. ∎

Lemma 3.4 *For $t \geq 6$, let $\lambda \in \mathcal{DS}(t)$ have the largest size. If there exists an index i such that $h_i, h_{i+1} \in MD(\lambda)$ and $d_i = 6$, then $d_j = 6$ for all $j > i$.*

Proof. Assume $\lambda \in \mathcal{DS}(t)$ achieves the largest size, where $t \geq 6$. Suppose $h_i, h_{i+1} \in MD(\lambda)$ with $d_i = 6$. By Lemma 3.2, assume for contradiction that there exists a smallest $q > i$ such that $h_q, h_{q+1} \in MD(\lambda)$ with $d_q = 4$. Then $d_{q-1} = 6$ and $h_{q-1}, h_q \in \mathrm{Q}_t^r$. By Lemma 3.3, we have $2t + 4 - h_{q-1} = 2t - 2 - h_q \in MD(\lambda)$. Define λ' by modifying $MD(\lambda)$ as follows:

- Replace h_q with $h_q + 2$,
- Replace $2t - 2 - h_q$ with $2t + 2 - h_q$.

One can easily derive that $MD(\lambda')$ satisfies the three conditions in Theorem 2.4 and has a larger sum of the elements than $MD(\lambda)$. Then we have $|\lambda'| > |\lambda|$, a contradiction. Hence, no such q exists. This completes the proof. ∎

Lemma 3.5 *For $t \geq 6$, let $\lambda \in \mathcal{DS}(t)$ have the largest size, with $d(\lambda) = (d_1, d_2, \ldots, d_{k-1})$, where $k \geq 2$. Suppose there exists an index $1 \leq \ell \leq k$ such that $d_{\ell-1} = 4$ and $d_\ell = 6$ (with conventions: $\ell = 1$ means all $d_i = 6$; $\ell = k$ means all $d_i = 4$). Then*

$$\lfloor \frac{2t+1}{12} \rfloor \leq \ell \leq \lfloor \frac{2t+1}{12} \rfloor + 2. \tag{3.1}$$

Proof. Suppose $\lambda \in \mathcal{DS}(t)$ achieves the largest size with $MD(\lambda) = \{h_1, h_2, \ldots, h_k\}$, where $t \geq 6$ and $k \geq 2$. Let $d(\lambda) = (d_1, d_2, \ldots, d_{k-1})$, where $d_{\ell-1} = 4$ and $d_\ell = 6$ for some $1 \leq \ell \leq k$. By Lemmas 3.2 and 3.4, we have $d_1 = \cdots = d_{\ell-1} = 4$ and $d_\ell = \cdots = d_{k-1} = 6$. We claim that $h_\ell \in \mathrm{Q}_t^r$. If not, then $h_\ell \in \mathrm{Q}_t^l$. From the structure of Q_t^r and Q_t^l, the minimum element of Q_t^r exceeds the maximum element of Q_t^l by 4, while their sum is either $2t$ or $2t+2$. Hence, by Theorem 2.4(iii), these two extremal elements cannot both lie in $MD(\lambda)$. This means that $d_{\ell-1} \geq 6$, contradicting the fact that $d_{\ell-1} = 4$.

Thus, $h_\ell \in \mathrm{Q}_t^r$. Combining these results with Theorem 2.4(i) and Lemma 3.1, we deduce that

$$h_i = \begin{cases} 2t+3-4i & \text{if } 1 \le i \le \ell, \\ 2t+3-4\ell-6(i-\ell) & \text{if } \ell+1 \le i \le k. \end{cases}$$

Since $t \ge 6$, we write $t = 6a+p$, where $a \ge 1$ and $0 \le p \le 5$. In what follows, we perform a detailed analysis of the possible range of ℓ.

We first establish the lower bound $\ell \ge \lfloor \frac{2t+1}{12} \rfloor$. Suppose for contradiction that $1 \le \ell \le \lfloor \frac{2t+1}{12} \rfloor - 1$. Then $1 \le \ell \le a-1$ and thus $a \ge 2$. Therefore,

$$h_\ell = 2t+3-4\ell \ge 2t+3-4\times\left(\lfloor \frac{2t+1}{12} \rfloor - 1\right) = 8a+2p+7. \tag{3.2}$$

By Lemma 3.3, we have $2t+4-h_\ell \in MD(\lambda)$. Next, we prove $2t+10-h_\ell \in MD(\lambda)$ by considering two cases.

(1) Suppose $h_\ell - 12 \in \mathrm{Q}_t^r$. Since $d_\ell = \cdots = d_{k-1} = 6$, we have $h_{\ell+1} = h_\ell - 6$ and $h_{\ell+2} = h_\ell - 12$. By Lemma 3.3, it then follows that $2t+4-h_{\ell+1} = 2t+10-h_\ell \in MD(\lambda)$.

(2) Suppose $h_\ell - 12 \in \mathrm{Q}_t^l$. From the structure of Q_t^r and Q_t^l, together with the facts that $d_{\ell+1} = 6$ and $h_{\ell+1} = h_\ell - 6 \in \mathrm{Q}_t^r$, we deduce that

$$h_{\ell+2} = h_\ell - 12 = \begin{cases} t-4 & \text{if } t \text{ is odd}, \\ t-1 & \text{if } t \text{ is even}. \end{cases}$$

However, if $h_\ell - 12 = t-4$, then

$$h_\ell = t+8 = 6a+p+8,$$

which is impossible by (3.2) together with $a \ge 2$ and $0 \le p \le 5$. Therefore, we must have $h_\ell - 12 = t-1$, with t even, and then $h_\ell - 12 = 2t+10-h_\ell$, so $2t+10-h_\ell \in MD(\lambda)$.

Define $\mu \in \mathcal{DS}(t)$ by modifying $MD(\lambda)$:

- Replace $h_{\ell+1}$ with $h_\ell - 4$,
- Replace $2t+10-h_\ell$ with $h_\ell - 8$,
- Delete $2t+4-h_\ell$.

Since λ has the largest size, we have $|\lambda| \geq |\mu|$. Applying $h_{\ell+1} = h_\ell - 6$ and Lemma 2.2, we obtain

$$|\lambda| - |\mu| = \big[(h_\ell - 6) + (2t + 4 - h_\ell) + (2t + 10 - h_\ell)\big] - \big[(h_\ell - 4) + (h_\ell - 8)\big] \geq 0,$$

which simplifies to

$$h_\ell \leq \frac{4t + 20}{3} = 8a + \frac{4p + 20}{3}.$$

This contradicts (3.2). Therefore, $\ell \geq \lfloor \frac{2t+1}{12} \rfloor$.

Now we prove the upper bound $\ell \leq \lfloor \frac{2t+1}{12} \rfloor + 2$. Assume for contradiction that $\ell \geq \lfloor \frac{2t+1}{12} \rfloor + 3$, so $\ell \geq a + 3$. Then

$$h_\ell = 2t + 3 - 4\ell \leq 2t + 3 - 4 \times (\lfloor \frac{2t + 1}{12} \rfloor + 3) = 8a + 2p - 9. \tag{3.3}$$

It follows that $h_\ell + 8 \leq 8a + 2p - 1 < 12a + 2p + 1 = 2t + 1$. Since $d_1 = \cdots = d_{\ell-1} = 4$, we conclude that $MD(\lambda)$ contains the elements $h_\ell + 4$ and $h_\ell + 8$. We then construct a partition $\mu \in \mathcal{DS}(t)$ by modifying $MD(\lambda)$:

- Replace $h_\ell + 8$ by $h_\ell + 6$,
- Replace $h_\ell + 4$ by $2t - 2 - h_\ell$,
- Add $2t - 8 - h_\ell$.

By the maximality of λ, we have $|\lambda| \geq |\mu|$. Applying Lemma 2.2, we deduce that

$$|\lambda| - |\mu| = \big[(h_\ell + 8) + (h_\ell + 4)\big] - \big[(h_\ell + 6) + (2t - 2 - h_\ell) + (2t - 8 - h_\ell)\big] \geq 0.$$

This simplifies to

$$h_\ell \geq \frac{4t - 16}{3} = 8a + \frac{4p - 16}{3}.$$

Since $0 \leq p \leq 5$, we have $\frac{4p-16}{3} > 2p + 9$, contradicting (3.3), which states that $h_\ell \leq 8a + 2p - 9$. Hence, we must have $\ell \leq \lfloor \frac{2t+1}{12} \rfloor + 2$, completing the proof. ∎

Now we are ready for the proof of Theorem 1.3.

Proof of Theorem 1.3. We characterize $MD(\lambda)$ for $\lambda \in \mathcal{DS}(t)$ with the largest size, where $t \geq 3$.

Let $MD(\lambda) = \{h_1, h_2, \ldots, h_k\}$ with $h_i > h_{i+1}$ for $1 \leq i \leq k - 1$. When $k = 1$ (i.e., $3 \leq t \leq 5$), Lemma 3.1 immediately gives $MD(\lambda) = \{2t-1\}$. Thus, by Lemma 2.2, we have $|\lambda| = 2t - 1$. When $k \geq 2$ (i.e., $t \geq 6$), let $d(\lambda) = (d_1, d_2, \ldots, d_{k-1})$, where $d_{\ell-1} = 4$

and $d_\ell = 6$ for some $1 \le \ell \le k$. By Lemma 3.5, we have $d_1 = d_2 = \cdots = d_{\lfloor \frac{2t+1}{12} \rfloor - 1} = 4$. By Lemma 3.1, we have $h_i = 2t + 3 - 4i$ for $1 \le i \le \lfloor \frac{2t+1}{12} \rfloor$, that is,

$$\{2t-1, 2t-5, \ldots, 2t+3-4\lfloor \frac{2t+1}{12} \rfloor\} \subseteq MD(\lambda).$$

It remains to determine ℓ precisely.

Without loss of generality, we assume that t is even and $t \ge 6$. Then $|\mathrm{Q}_t^r| = \frac{t}{2} - 1$. Note that $\{2t-1, 2t-5, \ldots, 2t+3-4\lfloor \frac{2t+1}{12} \rfloor\} \subseteq MD(\lambda)$. By the definition of partitions in $\mathcal{DS}(t)$, it follows that $2t-3, 2t-7, \ldots, 2t+1-4\lfloor \frac{2t+1}{12} \rfloor \notin MD(\lambda)$. Hence, we obtain

$$|\mathrm{Q}_t^r \backslash \{2t-1, 2t-3, \ldots, 2t+1-4\lfloor \frac{2t+1}{12} \rfloor\}| = \frac{t}{2} - 1 - 2\lfloor \frac{2t+1}{12} \rfloor,$$

which is denoted by Δ. We analyze based on $\Delta \mod 3$.

Case 1. $\Delta = \frac{t}{2} - 1 - 2\lfloor \frac{2t+1}{12} \rfloor \equiv \pmod 3$. Combining Lemmas 3.2, 3.3, 3.4, and 3.5, we derive $t+5, t-1 \in MD(\lambda)$ when $\ell \ne k$. Furthermore, we deduce that $2t-3-4\lfloor \frac{2t+1}{12} \rfloor, 2t-9-4\lfloor \frac{2t+1}{12} \rfloor, \ldots, t+5, t-1, \ldots, 7+4\lfloor \frac{2t+1}{12} \rfloor \in MD(\lambda)$. Recall that $2t+3-4\lfloor \frac{2t+1}{12} \rfloor \in MD(\lambda)$. Hence, by Lemma 3.3, it follows that $1+4\lfloor \frac{2t+1}{12} \rfloor \in MD(\lambda)$ guaranties that λ reaches the maximum possible size. Therefore, we deduce that $\ell = \lfloor \frac{2t+1}{12} \rfloor$, namely,

$$d(\lambda) = (d_1, \ldots, d_{\lfloor \frac{2t+1}{12} \rfloor - 1}, d_{\lfloor \frac{2t+1}{12} \rfloor}, \ldots, d_{k-1}) = (4, \ldots, 4, 6, \ldots, 6).$$

Hence, we have

$$MD(\lambda) = \{2t-1, 2t-5, \ldots, 2t+3-4\lfloor \frac{2t+1}{12} \rfloor, 2t-3-4\lfloor \frac{2t+1}{12} \rfloor, \ldots, 1+4\lfloor \frac{2t+1}{12} \rfloor\}.$$

According to Lemma 2.2, the size of λ is given by

$$|\lambda| = \frac{1}{3}(t-1)\left(t+1-4\lfloor \frac{2t+1}{12} \rfloor\right) + \lfloor \frac{2t+1}{12} \rfloor \left(2t+1-2\lfloor \frac{2t+1}{12} \rfloor\right)$$

as desired.

For example, let $t = 24$ and λ be a partition in $\mathcal{DS}(24)$ with the largest size. Then $47, 43, 39, 35 \in MD(\lambda)$. Furthermore, we have $\ell = \lfloor \frac{2\times 24+1}{12} \rfloor = 4$ and $\Delta = \frac{24}{2} - 1 - 2 \times 4 = 3$. It follows that $29, 23, 17 \in MD(\lambda)$. Hence, we obtain $MD(\lambda) = \{47, 43, 39, 35, 29, 23, 17\}$ as shown in Fig. 5. Consequently, the size of λ is $|\lambda| = 233$.

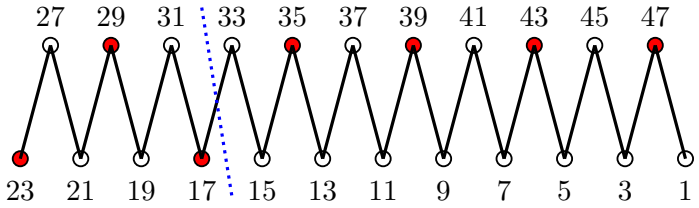


Fig. 5: The representation of $MD(\lambda) = \{47, 43, 39, 35, 29, 23, 17\}$.

Case 2. $\Delta = \frac{t}{2} - 1 - 2\lfloor \frac{2t+1}{12} \rfloor \equiv 1 \pmod 3$. Combining Lemmas 3.2, 3.3, 3.4, and 3.5, we derive that $t+5, t-1 \in MD(\lambda)$ when $\ell \neq k$. This further implies that $\Delta \geq 4$; indeed, if $\Delta = 1$, then $t+3 \in MD(\lambda)$ and $\ell = k$, which corresponds to the case $d(\lambda) = (d_1, d_2, \ldots, d_{k-1}) = (4, 4, \ldots, 4)$. Moreover, we obtain $2t-5-4\lfloor \frac{2t+1}{12} \rfloor, 2t-11-4\lfloor \frac{2t+1}{12} \rfloor, \ldots, t+5, t-1, \ldots, 9+4\lfloor \frac{2t+1}{12} \rfloor \in MD(\lambda)$. Since $2t+3-4\lfloor \frac{2t+1}{12} \rfloor \in MD(\lambda)$, to ensure that λ achieves the maximum size, we must also include $2t-1-4\lfloor \frac{2t+1}{12} \rfloor \in MD(\lambda)$. Consequently, we have $\ell = \lfloor \frac{2t+1}{12} \rfloor + 2$, and thus

$$d(\lambda) = (d_1, \ldots, d_{\lfloor \frac{2t+1}{12} \rfloor + 1}, d_{\lfloor \frac{2t+1}{12} \rfloor + 2}, \ldots, d_{k-1}) = (4, \ldots, 4, 6, \ldots, 6).$$

Hence, we have

$$MD(\lambda) = \{2t-1, 2t-5, \ldots, 2t-5-4\lfloor \frac{2t+1}{12} \rfloor, 2t-11-4\lfloor \frac{2t+1}{12} \rfloor, \ldots, 9+4\lfloor \frac{2t+1}{12} \rfloor\}.$$

By Lemma 2.2, the size of λ is given by

$$|\lambda| = \frac{1}{3}(t-1)\left(t-7-4\lfloor \frac{2t+1}{12} \rfloor\right) + \left(\lfloor \frac{2t+1}{12} \rfloor + 2\right)\left(2t-3-2\lfloor \frac{2t+1}{12} \rfloor\right)$$

as desired.

For example, let $t = 26$ and λ be a partition in $\mathcal{DS}(26)$ with the largest size. Then $51, 47, 43, 39 \in MD(\lambda)$. Moreover, we have $\ell = \lfloor \frac{2\times 26+1}{12} \rfloor + 2 = 6$ and $\Delta = \frac{26}{2} - 1 - 2 \times 4 = 4$. It follows that $35, 31, 25 \in MD(\lambda)$. Hence, we obtain $MD(\lambda) = \{51, 47, 43, 39, 35, 31, 25\}$ as shown in Fig. 6. Consequently, the size of λ is $|\lambda| = 271$.

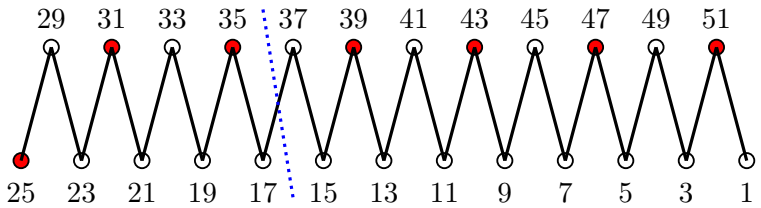


Fig. 6: The representation of $MD(\lambda) = \{51, 47, 43, 39, 35, 31, 25\}$.

Case 3. $\Delta = \frac{t}{2} - 1 - 2\lfloor\frac{2t+1}{12}\rfloor \equiv 2 \pmod 3$. Similarly, combining Lemmas 3.2, 3.3, 3.4, and 3.5, we deduce that $t+5, t-1 \in MD(\lambda)$ when $\ell \neq k$. Furthermore, we derive $2t-1-4\lfloor\frac{2t+1}{12}\rfloor, 2t-7-4\lfloor\frac{2t+1}{12}\rfloor, \ldots, t+5, t-1, \ldots, 5+4\lfloor\frac{2t+1}{12}\rfloor \in MD(\lambda)$. Since $2t+3-4\lfloor\frac{2t+1}{12}\rfloor \in MD(\lambda)$, it follows that $\ell = \lfloor\frac{2t+1}{12}\rfloor + 1$, and consequently

$$d(\lambda) = (d_1, \ldots, d_{\lfloor\frac{2t+1}{12}\rfloor}, d_{\lfloor\frac{2t+1}{12}\rfloor+1}, \ldots, d_{k-1}) = (4, \ldots, 4, 6, \ldots, 6).$$

Hence, we have

$$MD(\lambda) = \{2t-1, 2t-5, \ldots, 2t-1-4\lfloor\frac{2t+1}{12}\rfloor, 2t-7-4\lfloor\frac{2t+1}{12}\rfloor, \ldots, 5+4\lfloor\frac{2t+1}{12}\rfloor\}.$$

By Lemma 2.2, the size of λ is given by

$$|\lambda| = \frac{1}{3}(t-1)\left(t-3-4\lfloor\frac{2t+1}{12}\rfloor\right) + \left(\lfloor\frac{2t+1}{12}\rfloor + 1\right)\left(2t-1-2\lfloor\frac{2t+1}{12}\rfloor\right)$$

as desired.

For example, let $t = 28$ and λ be a partition in $\mathcal{DS}(28)$ with the largest size. Then $55, 51, 47, 43 \in MD(\lambda)$, Moreover, we have $\ell = \lfloor\frac{2\times 28+1}{12}\rfloor + 1 = 5$ and $\Delta = \frac{28}{2} - 1 - 2\times 4 = 5$. It follows that $39, 33, 27, 21 \in MD(\lambda)$. Hence, we obtain $MD(\lambda) = \{55, 51, 47, 43, 39, 33, 27, 21\}$ as shown in Fig. 7. Consequently, the size of λ is $|\lambda| = 316$.

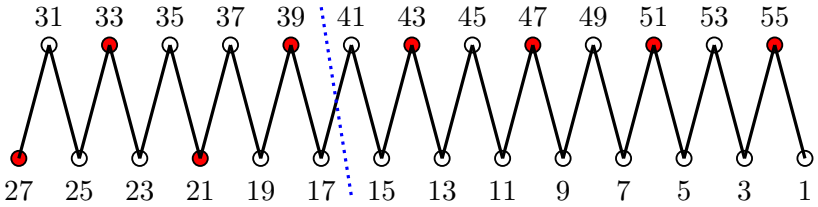


Fig. 7: The representation of $MD(\lambda) = \{55, 51, 47, 43, 39, 33, 27, 21\}$.

When t is odd and $t \geq 6$, one can verify that $|Q_t^r| = \frac{t-1}{2}$. Recall that $\{2t-1, 2t-5, \ldots, 2t+3-4\lfloor\frac{2t+1}{12}\rfloor\} \subseteq MD(\lambda)$. Then, by the definition of partitions in $\mathcal{DS}(t)$, we have $2t-3, 2t-7, \ldots, 2t+1-4\lfloor\frac{2t+1}{12}\rfloor \notin MD(\lambda)$. Thus,

$$|Q_t^r \backslash \{2t-1, 2t-3, \ldots, 2t+1-4\lfloor\frac{2t+1}{12}\rfloor\}| = \frac{t-1}{2} - 2\lfloor\frac{2t+1}{12}\rfloor.$$

Similarly, we need to consider three cases: (i) $\frac{t-1}{2} - 2\lfloor\frac{2t+1}{12}\rfloor \equiv 0 \pmod 3$; (ii) $\frac{t-1}{2} - 2\lfloor\frac{2t+1}{12}\rfloor \equiv 1 \pmod 3$; and (iii) $\frac{t-1}{2} - 2\lfloor\frac{2t+1}{12}\rfloor \equiv 2 \pmod 3$. These three cases can be handled in a similar manner to the case of even t and are therefore omitted.

In all cases, $MD(\lambda)$ is uniquely determined for a partition of maximum size. Hence, there is exactly one such partition, completing the proof of Theorem 1.3. ∎

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References

[1] T. Amdeberhan, Theorems, problems and conjectures, preprint, `arXiv:1207.4045`.

[2] T. Amdeberhan and E. Leven, Multi-cores, posets, and lattice paths, *Adv. in Appl. Math.*, **71** (2015), 1-13.

[3] J. Anderson, Partitions which are simultaneously t_1- and t_2-core, *Discrete Math.*, **248** (2002), 237-243.

[4] G.E. Andrews and K. Eriksson, *Integer Partitions*, Cambridge University Press, 2004.

[5] D. Armstrong, C.R.H. Hanusa, and B. Jones, Results and conjectures on simultaneous core partitions, *European J. Combin.*, **41** (2014), 205-220.

[6] W.Y.C. Chen, H.H.Y. Huang, and L.X.W. Wang, Average size of a self-conjugate (s,t)-core partition, *Proc. Amer. Math. Soc.*, **144(4)** (2016), 1391-1399.

[7] H. Cho, J. Huh, and J. Sohn, Counting self-conjugate $(s, s+1, s+2)$-core partitions, *Ramanujan J.*, **55** (2021), 163-174.

[8] H. Cho, J. Huh, and J. Sohn, The $(s, s+d, \cdots, s+pd)$-core partitions and the rational Motzkin paths, *Adv. in Appl. Math.*, **121** (2020), 102096.

[9] H. Cho and J. Huh, Self-conjugate $(s, s+d, \cdots, s+pd)$-core partitions and free Motzkin paths, *Ramanujan J.*, **57(3)** (2022), 907-929.

[10] H. Cho, B. Kim, H. Nam, J. Sohn, A survey on t-core partitions, *Hardy-Ramanujan J.*, **44** (2022), 81-101.

[11] H. Cho and K. Hong, Corners of self-conjugate $(s, s+1)$-cores and $(\overline{s}, \overline{s+1})$-cores, *Discrete Math.*, **345(9)** (2022), 112949.

[12] C. Even-Zohar, Sizes of simultaneous core partitions, *J. Comb. Theory Ser. A.*, **185** (2022), 105536.

[13] B. Ford, H. Mai, and L. Sze, Self-conjugate simultaneous p- and q-core partitions and blocks of A_n, *J. Number Theory*, **129(4)** (2009), 858-865.

[14] H.H.Y. Huang and L.X.W. Wang, The corners of core partitions, *SIAM J. Discrete Math.*, **32(3)** (2018), 1887–1902.

[15] P. Johnson, Lattice points and simultaneous core partitions, *Electron. J. Combin.*, **25(3)** (2018), Paper 3.47.

[16] W. J. Keith and R. Nath, Partitions with prescribed hooksets, *J. Comb. Number Theory.*, **3(1)** (2011) 39–50.

[17] N. Kravitz, On the number of simultaneous core partitions with d-distinct parts, *Discrete Math.*, **342(12)** (2019), 111592.

[18] W. Ma and X. Jiang, On the largest sizes of $(s, qs \pm 1)$-core partitions with parts of the same parity, *Ramanujan J.*, **61** (2023), 895-908.

[19] I.G. Macdonald, *Symmetric Functions and Hall Polynomials*, Oxford Mathematical Monographs, The Clarendon Press, Oxford University Press, New York, second edition, 1995.

[20] H. Nam and M. Yu, The largest size of an $(s, s+1)$-core partition with parts of the same parity, *Int. J. Number Theory*, **17(3)** (2021), 699-712.

[21] R. Nath, Symmetry in maximal $(s-1, s+1)$ cores, *Integers*, **16** (2016), Paper No. A18.

[22] R. Nath and J.A. Sellers, Abaci structures of $(s, ms \pm 1)$-core partitions, *Electron. J. Combin.*, **24(1)** (2017), Paper 1.5.

[23] J. Olsson and D. Stanton, Block inclusions and cores of partitions, *Aequationes Math.*, **74(1-2)** (2007), 90-110.

[24] K. Paramonov, Cores with distinct parts and bigraded Fibonacci numbers, *Discrete Math.*, **341(4)** (2018), 875-888.

[25] R.P. Stanley, *Enumerative Combinatorics*, vol. 2, Cambridge University Press, New York/Cambridge, 1999.

[26] R.P. Stanley and F. Zanello, The Catalan case of Armstrong's conjectures on simultaneous core partitions, *SIAM J. Discrete Math.*, **29(1)** (2015), 658-666.

[27] A. Straub, Core partitions into distinct parts and an analog of Euler's theorem, *European J. Combin.*, **57** (2016), 40-49.

[28] V. Wang, Simultaneous core partitions: parameterizations and sums, *Electron. J. Combin.*, **23(1)** (2016), Paper 1.4.

[29] H. Xiong, Core partitions with distinct parts, *Electron. J. Combin.*, **25(1)** (2018), Paper 1.57.

[30] H. Xiong, On the largest sizes of certain simultaneous core partitions with distinct parts, *European J. Combin.*, **71** (2018), 33-42.

[31] H. Xiong and W.J.T. Zang, On the polynomiality and asymptotics of moments of sizes for random $(n, dn \pm 1)$-core partitions with distinct parts, *Sci. China Math.*, **64** (2021), 869-886.

[32] S.H.F. Yan, D. Yan, and H. Zhou, Self-conjugate $(s, s+d, s+2d)$-core partitions and free Motzkin paths, *Discrete Math.*, **344(4)** (2021), 112304.

[33] S.H.F. Yan, Y. Yu, and H. Zhou, On self-conjugate $(s, s+1, \cdots, s+k)$-core partitions, *Adv. in Appl. Math.*, **113** (2020), 101975.

[34] A. Zaleski, Explicit expressions for the moments of the size of an $(s, s+1)$-core partition with distinct parts, *Adv. in Appl. Math.*, **84** (2017), 1-7.

[35] A. Zaleski, Explicit expressions for the moments of the size of an $(n, dn-1)$-core partition with distinct parts, *Integers*, **19** (2019), p1.

[36] A. Zaleski and D. Zeilberger, Explicit expressions for the expectation, variance and higher moments of the size of a $(2n+1, 2n+3)$-core partition with distinct parts, *J. Difference Equ. Appl.*, **23(7)** (2017), 1241–1254.